NEERAJ NAGA MANIKANTA PRAKASH PADARTHI

UAV & Mechanical Design Engineer | Defense Aerospace | Structural Analysis | R&D
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Professional Summary

UAV Design Engineer with 2.5+ years of hands-on experience in fixed-wing, tandem-wing, and multirotor platforms for Indian defense and research applications. Skilled in aerospace CAD modeling (CATIA, SolidWorks, NX Siemens), structural and aerodynamic analysis using FEA (NX Nastran, ANSYS Mechanical) and CFD tools, topology optimization, and lightweight composite airframe design. Proven track record across IIT Kanpur R&D lab and defense UAV production environments — from concept and CAD through prototype, flight validation, and delivery. Co-PI on a funded **Rs. 21L IIT Roorkee** project; **4 Patent Applicant – Multiple Innovations**; **ISRO internship** alumnus.

Education Background

- 2024 – B.Tech Mechanical Engineering, GMR Institute of Technology.
- 2020 – Intermediate (MPC), Narayana Junior College.
- 2017 – Secondary School Education, Ravindra Bharathi School.

Engineering Experience

- **Senior Design Engineer, Magnum Wings Pvt Ltd** *11/2025 – Present*
  - **Heavy-Lift Cargo UAV Structural Design (25 kg Payload Platform)**
    - Led full structural and aerodynamic design of a 25 kg payload cargo UAV using CATIA, from concept to prototype-ready manufacturing drawings.
    - Designed and optimized aerodynamic control systems including flaps and control surfaces, improving control authority and responsiveness across the full flight envelope.
    - Developed self-locking rib internal architecture, increasing wing torsional stiffness while simultaneously reducing part count by ∼20%, improving both performance and manufacturability.
    - Applied topology optimization to key airframe load-bearing components, achieving all structural targets with minimized material usage and weight.
    - Coordinated CAD development, prototype fabrication, and cross-functional integration with manufacturing teams across 3+ subsystems from design freeze to first article.
  - **Heavy-Lift Cargo UAV Platform Development (100 kg Payload)**
    - Contributed to structural design and development of a 100 kg payload heavy-lift UAV – one of the highest payload-class platforms in Indian private UAV development.
    - Designed UAV landing gear system from scratch: load path definition, structural support members, and shock absorption sizing for hard-landing load cases up to 3g.
    - Performed FEA-based structural validation and iterative design optimization across multiple load cases to meet airworthiness and operational load requirements.
    - Supported full airframe structural architecture definition and multi-subsystem integration for large-format cargo UAV configuration.
- **R&D Facilitated UAV Lab, IIT Kanpur – VU Dynamics Pvt Ltd** *05/2024 – 10/2025*
  - **Comprehensive R&D – UAV Systems Developments**
    - Served as Technical In-Charge for design and manufacturing of 6+ UAV platforms across fixed-wing, multirotor, and hybrid categories at a premier IIT-facilitated defense R&D lab.
    - Led end-to-end design and production of fixed-wing UAVs: 20 kg flying platforms, warhead-equipped drones, single-grenade release mechanisms, Quadcopter and FPV drones – all delivered within project timelines and budget.
    - Oversaw manufacturing and assembly of Tandem-wing (20 kg), flying wings, and multi-weight drones (12 kg, 15 kg with 2.5 kg warhead payload quadcopter) with full structural and flight validation.
    - Created 50+ detailed CAD models for UAV components, jigs, tools, and gauges; validated all designs through FEM analysis prior to fabrication, eliminating costly rework.
    - Contributed directly to Indian defense UAV prototypes – from rapid design iteration through subsystem integration, field testing, and flight validation in operational conditions.
  - **Flying Wing Full Design with Jigs and Fixtures (Fixed-Wing UAV)**
    - Led complete flying-wing UAV design in SolidWorks – airframe, control surfaces, and structural components – achieving full manufacturability sign-off validated through wind tunnel testing and CFD-based aerodynamic analysis.
    - Designed a complete jig-and-fixture set enabling highly repeatable UAV assembly with dimensional tolerance under ±0.5 mm, supporting batch production of multiple units simultaneously.
    - Eliminated warping and misalignment issues present in previous builds, reducing rework time by ∼30% and significantly improving overall production throughput and consistency.

Engineering Experience / Internship

- **Internship & Project Trainee, ISRO – LEOS Bangalore** *2023*
  - Worked on the 6-DoF hexapod (Stewart platform), a parallel manipulator capable of precise movement in all six spatial degrees of freedom – translation (X, Y, Z) and rotation (roll, pitch, yaw) – used for precision motion simulation and calibration.
  - Contributed to mechanical sensor mount designs and FEM analysis for ROSAT-3 satellite subsystems using NX Siemens; ensured structural integrity under launch and operational vibration load cases.
  - Gained hands-on experience in NX Siemens CAD modeling, modal analysis, and structural simulations for space-grade hardware applications.
  - Trained in opto-mechanical design principles, with practical focus on precision alignment techniques using autocollimators for high-accuracy optical systems.

Hard Skills

- **CAD / Design Tools:** CATIA V5, SolidWorks, NX Siemens, AutoCAD
- **Simulation & Analysis:** FEA (NX Nastran, ANSYS Mechanical), CFD (ANSYS Fluent – basic), Structural & Modal Analysis, Topology Optimization, GD&T

- **UAV & Aerospace:** Fixed-Wing, Multirotor, Flying Wing, Tandem-Wing Design; Airframe Structural Architecture; Control Surface Design; Landing Gear Design; Payload Integration; Composite Airframe (CFRP/GFRP); Jigs & Fixtures
- **Manufacturing & Prototyping:** Rapid Prototyping, Sheet Metal, Composite Layup, CNC Machining Awareness, Manufacturing DFM/DFA
- **R&D & Management:** End-to-End Product Development, Design Validation, Flight Test Integration, Project & Team Management, Technical Documentation
- **Soft Tools:** MS Office, LaTeX, Python (basic scripting), Git (version control)

Achievements

- **Gold Medal – International Mathematics Olympiad (IMO):** City Rank: 1 | State Rank: 12 | International Rank: 24
- **Silver Medal – National Science Olympiad (NSO):** City Rank: 3 | International Rank: 201
- **Finalist – APSCHE Student Innovation Award 2024** – Recognized for outstanding innovation in engineering design and prototyping among state-level entries.
- **Winner – Idea Competition 2022, EDC GMRIT** – Secured 1st place for presenting a novel and feasible technological solution; organized by MSME & EDC Cell.
- **Certificate of Appreciation – Surya Supply & Transport Symposium, HQ Central Command, Indian Army** – Recognized by Major General S. T. A. Rizvi, MG ASC, Central Command, Lucknow (18 August 2025).
- **Defense UAV Project Contributor – Industry R&D** – Key role in design and production of fixed-wing and tandem-wing UAVs for Indian defense applications, including payload integration and full structural validation.
- **Patent Holder – 4 Published Patents** – AI-based smart lawn mower, thermoelectric cooling/heating flask, solar-powered thermoelectric refrigerator, and lubricant purification system (LPVD-DWFT).
- **ISRO Internship Completion – LEOS Bangalore** – Successfully completed ROSAT-3 sensor mount FEA and Stewart platform design; gained expertise in NX Siemens and opto-mechanical alignment.
- **Zonal-Level Football Player – JNTU-K** – Represented GMRIT at inter-collegiate zonal football tournaments (2020–2024).

Roles and Leadership

- **JNTUK Zonal Football Player** (2020–2024) – Represented GMRIT at zonal-level football tournaments under JNTU-K.
- **Student Chairperson, Robotics Club, GMRIT** (2020–2024) – Led robotics innovation initiatives and managed student projects, workshops, and competitions within the Centre of Excellence for Robotics & Automation.
- **Student Coordinator, SAE GMRIT** (2020–2024) – Coordinated student participation in SAEINDIA events, fostering technical exposure through competitions, workshops, and technical fests.
- **RJ, GMRIT Community Radio 90.4** (2020–2024) – Hosted and curated engaging science and innovation content to promote technical awareness among students and the public.
- **Team Lead, NSS Unit, GMRIT** (2020–2024) – Organized social service activities including rural development camps, blood donation drives, and environmental awareness events.

Publications & Patents

Publications:

- *Design and Fabrication of Fully Automated Lawn Mower* – Published in IJRPR. Focuses on development of a cost-effective, autonomous lawn mowing system integrating mechanical design with control systems.

Patents:

- **AI-Based Smart Lawn Mower** – Fully autonomous, AI-integrated lawn mower with smart navigation, solar self-charging, and multi-mode operation.
- **Instantaneous Cooling and Heating Liquid Flask** – Smart thermoelectric flask for rapid liquid temperature control using copper and carbon fiber with smart regulation.
- **Solar Powered Thermoelectric Refrigerator** – Compact, energy-efficient off-grid refrigeration system powered by solar panels and TEC modules.
- **Lubricant Purification System using LPVD-DWFT** – Vacuum dehydration-based system for removing water, dust, and contaminants from lubricants.

Projects & Research Highlights

- **MJESTO – A Conceptual Space Habitat** – Futuristic space town design for long-term human survival beyond Earth; focused on sustainability, modularity, and adaptability in extraterrestrial environments.
- **Natural Fiber Composites Optimization** – Analyzed mechanical and thermal properties of eco-friendly composites using banana fiber, coconut midrib, and rice husk; explored flame-retardant additives for construction, automotive, and electronics sectors.
- **AWM – Atmospheric Water Management System** – Sustainable irrigation solution for arid regions using atmospheric water vapor condensation; aimed at reducing water scarcity and deforestation.
- **Aluminum Metal Matrix Composites (MMC)** – Developed Al-Mg based MMCs using Friction Stir Processing (FSP) for aerospace and automotive applications; focused on enhancing mechanical strength, weight reduction, and thermal resistance.
- **Miniature Hexapod (Stewart Platform) – ISRO, LEOS Bangalore** – Design and analysis of a 6-DoF Stewart platform for precision motion simulation; NX Siemens structural and modal analysis.
- **ROSAT-3 Satellite Sensor Mounts – ISRO, LEOS Bangalore** – FEA and design validation of satellite sensor mounts; opto-mechanical precision engineering and alignment techniques.

Funded Projects, Trainings & Conferences

- **AI-SMS Lawn Mower (G-LAWN)** – Co-PI, IIT Roorkee iHub **Rs. 21 Lakhs** funded project.
- **ICHT Flask** – GMRIT Innovation Fund Rs. 12,000 funded prototype development.
- **UNHCR IMUN** Online – Delegate of Vietnam, scored 88.0%.
- **CII & ISRO** International Conference on Space, 2023 – Presented and participated as delegate.

Languages

- **Telugu** – Native | **English** – Professional proficiency | **Hindi** – Conversational proficiency